Employee Success Analytics at NextGen Corp.

BUSINESS OVERVIEW

NextGen Corp. is a growing technology company focused on developing innovative solutions in the software and hardware spaces. The company prides itself on attracting top talent and maintaining high employee satisfaction to drive growth.

PROBLEM STATEMENT / AIM OF PROJECT

There are increasing concerns regarding employee turnover, performance variability, and salary disparities within departments.

To ensure continued success, NextGen Corp. needs to optimize employee retention, track employee performance consistently and maintain fair salary structures across departments. The HR department needs a data-driven approach to:

- Identify trends and patterns in employee retention and turnover.
- Track and evaluate performance across different departments.
- Assess the relationship between salary and performance to ensure fairness and employee satisfaction.

EMPLOYEE RETENTION ANALYSIS

Top 5 Highest Serving Employees

```
--Q1. Who are the top 5 highest serving employees?
SELECT
    DISTINCT(employee_id),
    first_name ||' ' || last_name AS Fullname,
    hire_date,
    job_title,
    current_date - hire_date as "Total days of service"
FROM employee
WHERE employee_id NOT IN (SELECT DISTINCT employee_id FROM turnover)
ORDER BY hire_date
LIMIT 5;
```

David Moore is the highest serving employee having served for 3654 days which is equivalent to about 10 years. 4 out of the top 5 serving employees are from the sales department.

	employee_id [PK] integer	fullname text	hire_date date	job_title character varying (30)	Total days of service integer
1	8	David Moore	2015-06-30	Sales Representative	3654
2	44	John Johnson	2015-10-27	Sales Manager	3535
3	11	Frank Johnson	2016-06-12	Sales Manager	3306
4	26	Jane Lee	2016-07-19	HR Specialist	3269
5	99	Eve Wilson	2016-10-13	Sales Representative	3183

Turnover Rate for Each Department

```
--Q2. What is the turnover rate for each department?
SELECT
    d.department_name,
    COUNT(DISTINCT(t.turnover_id)) AS Total_turnover,
    ROUND(COUNT(t.turnover_id) * 100.0/ COUNT(e.employee_id), 2) || '%' AS turnover_rate
FROM department d
LEFT JOIN employee e ON e.department_id = d.department_id
LEFT JOIN turnover t ON e.employee_id = t.employee_id
GROUP BY d.department_name
ORDER BY turnover_rate DESC;
```

	department_name character varying (30)	total_turnover bigint	turnover_rate text
1	Marketing	13	92.86%
2	Engineering	4	66.67%
3	Sales	8	27.59%
4	HR	3	27.27%

Marketing Department has the highest turnover rate of 92.86%. This may possibly be due to not been satisfied with salary as the average salary of the marketing department is $77857 compared to the company's average of $80,833.

Sales and HR has the lowest turnover rate of 27.59% and 27.27% respectively. The highest serving employees are also from this two departments which indicates the employees are more satisfied. It is worth noting that sales rep and HR specialist has the highest average salaries.

Employees at Risk of Leaving Based on Their Performance

```sql
--Q3. Which employees are at risk of leaving based on their performance?
SELECT
    DISTINCT e.employee_id,
    e.first_name  || ' ' || e.last_name AS Fullname,
    e.job_title,
    ROUND(AVG(p.performance_score), 2) AS "Average Performace Score",
    CASE
        WHEN AVG(p.performance_score) < 3.5 THEN 'High Risk of leaving'
        when avg (p.performance_score) BETWEEN 3.5 and 3.8 THEN 'Moderate Risk'
        ELSE 'No risk'
    END AS "Risk of Leaving based on Performance"
FROM employee e
JOIN performance p ON p.employee_id = e.employee_id
GROUP BY 1,2,3
ORDER BY "Average Performace Score";
```

Grace Wilson is the employee with the highest risk of leaving with an average performance score of 3.43 out of 5.

Employees with moderate risk of leaving seem to be within the same bracket in terms of performance score.

	employee_id [PK] integer	fullname text	job_title character varying (30)	Average Performace Score numeric	Risk of Leaving based on Performance text
1	22	Grace Wilson	Sales Representative	3.43	High Risk of leaving
2	1	Jane Wilson	Sales Manager	3.58	Moderate Risk
3	12	Eve Davis	HR Specialist	3.65	Moderate Risk
4	97	David Smith	Engineer	3.72	Moderate Risk
5	16	Charlie Lee	Sales Manager	3.77	Moderate Risk
6	40	Alice Wilson	HR Specialist	3.77	Moderate Risk
7	42	Alice Brown	Marketing Specialist	3.80	Moderate Risk
8	21	Grace Smith	Sales Manager	3.83	No risk

Main Reasons Employees are Leaving the Company

39.29% of employees who left were for personal reasons. This could be due to poor work-life balance, lower compensation rate etc. 25% of employees also left for another job.

```sql
--Q4. What are the main reasons employees are leaving the company
SELECT
    reason_for_leaving,
    COUNT(*) AS Total_Leavers,
    ROUND(COUNT(turnover_id) * 100.0 /
        (SELECT COUNT(*) FROM turnover),2) || '%' AS percentage_leavers
FROM turnover
GROUP BY reason_for_leaving
ORDER BY total_leavers DESC;
```

	reason_for_leaving text	total_leavers bigint	percentage_leavers text
1	Personal	11	39.29%
2	Found Another Job	7	25.00%
3	Career Growth	5	17.86%
4	Retired	5	17.86%

Recommendation

- Leverage on the experience and knowledge of long serving staff to train and mentor newer staff
- Offer flexible working arrangements to support relocation, childcare, or caregiving needs as this may reduce the number of employees leaving for personal reasons.
- Reassess campaign schedules, team size, and role clarity to avoid overloading top performers

EMPLOYEE PERFORMANCE ANALYSIS

How Many Employees Has Left the Company

```sql
--Q5. How many employees has left the company?
SELECT COUNT(DISTINCT(employee_id)) AS total_turnover
FROM turnover;
```

	total_turnover bigint
1	28

	department_name character varying (30)	total_turnover bigint	turnover_rate text
1	Marketing	13	92.86%
2	Engineering	4	66.67%
3	Sales	8	27.59%
4	HR	3	27.27%

A total of 28 employees representing about 47% of total employees have left the company between 2015 and 2025 . Out of the 28 employees, 13 of them are from the marketing department which was earlier discovered may be due to lower compensation rate.

How Many Employees Have a Performance Score of 5.0 / Below 3.5?

```
--Q6. How many employees have a performance score of 5.0 / below 3.5?
SELECT COUNT(DISTINCT(employee_id)) AS "Performance of 5.0"
FROM performance
WHERE performance_score = 5.0;

SELECT COUNT(DISTINCT(employee_id)) AS "Below 3.5 performance"
FROM performance
WHERE performance_score < 3.5;
```

	Below 3.5 performance bigint
1	45

	Performance of 5.0 bigint
1	9

45 out of 60 employees have a score of below 3.5 whereas 9 of the total employees have 5.0 score

Which Department Has The Most Employees With a Performance of 5.0 / Below 3.5?

```sql
--Q7. Which department has the most employees with a performance of 5.0 / below 3.5?
SELECT d.department_name, COUNT(DISTINCT(p.employee_id)) AS "Performance of 5.0"
FROM department d
LEFT JOIN performance p ON d.department_id = p.department_id
WHERE p.performance_score = 5.0
GROUP BY 1
ORDER BY "Performance of 5.0" DESC
LIMIT 1;

SELECT d.department_name, COUNT(DISTINCT(p.employee_id)) AS "Below 3.5 performance"
FROM department d
LEFT JOIN performance p ON d.department_id = p.department_id
WHERE p.performance_score < 3.5
GROUP BY 1
ORDER BY "Below 3.5 performance" DESC
LIMIT 1;
```

	department_name character varying (30)	Performance of 5.0 bigint
1	Marketing	5

	department_name character varying (30)	Below 3.5 performance bigint
1	Engineering	16
2	Marketing	16

Although marketing department has the most employees with performance of 5.0, the department also has the most employees with a score of below 3.5. This implies that a few of the employees are doing majority of the work which could result in stress and burnout leading to their decision to leave the company. The Engineering department also has equal number of employees with a performance of below 3.5

What is the Average Performance Score by Department?

```sql
--Q7. What is the average performance score by department?
SELECT d.department_name, ROUND(AVG(p.performance_score), 2) AS Average_Performance
FROM department d
LEFT JOIN performance p ON d.department_id = p.department_id
GROUP BY 1
ORDER BY Average_Performance DESC;
```

Although all departments have an average score of 4.0 and above, the marketing department tops all with a score of 4.13. This could be as a result of the department having the most employees with performance score of 5.0

	department_name character varying (30)	average_performance numeric
1	Marketing	4.13
2	Engineering	4.10
3	HR	4.05
4	Sales	4.00

Recommendation

- Use 1:1s to identify roadblocks and provide timely support to eliminate bottlenecks that prevent them from working efficiently
- Equip managers with coaching and leadership skills as good managers directly impact performance and retention
- Provide access to courses, certifications, and workshops relevant to each role

SALARY ANALYSIS

What is the Total Salary Expense for the Company

```
--Q8. What is the total salary expense for the company?
SELECT TO_CHAR(SUM(salary_amount), '$FM999,999,999') AS total_salary_expense
FROM salary;
```

	total_salary_expense text
1	$4,850,000

The company's average salary per employee is $80,833. The total salary paid for May 2024 alone stands at $4,850,000 - this is presumably the average of total salary paid monthly

```
--Average Employee Salary
SELECT
    TO_CHAR(AVG(salary_amount), '$FM999,999,999') AS average_employee_salary
FROM salary;
```

	average_employee_salary text
1	$80,833

What is the Average Salary by Job Title?

```sql
--Q9. What is the average salary by job title?
SELECT e.job_title, TO_CHAR(AVG(s.salary_amount), '$FM999,999.00') AS Average_salary
FROM employee e
LEFT JOIN salary s ON e.employee_id = s.employee_id
GROUP BY e.job_title
ORDER BY  Average_salary DESC;
```

The average salary of the various roles are $80,000 and above except marketing specialist which has an average salary of about $77,000. Although sales rep has the highest average salary, the employee with the highest risk of leaving is in this role which implies factors other than remuneration maybe contributing to employees in this role leaving the company.

	job_title character varying (30)	average_salary text
1	Sales Representative	$84,285.71
2	HR Specialist	$81,818.18
3	Engineer	$80,000.00
4	Sales Manager	$80,000.00
5	Marketing Specialist	$77,857.14

How Many Employees Earn Above 80,000

```sql
SELECT COUNT(DISTINCT(employee_id)) AS Employees_earning_above_80000
FROM salary
WHERE salary_amount > 80000
```

	employees_earning_above_80000 bigint
1	26

26 out of 60 employees (representing about 43% of employees) earn above $80,000 which implies that more than 50% earn below the company’s average employee salary of 80,833 which may indicate a general dissatisfaction with remuneration.

How Does Performance Correlate With Salary Across Departments?

```sql
SELECT
    d.department_name,
    Round(AVG(p.performance_score), 2) As "Average score",
    Round(AVG(s.salary_amount), 2) As "Average Salary Amount"
FROM
    employee e
LEFT JOIN department d ON e.department_id = d.department_id
LEFT JOIN salary s ON s.employee_id = e.employee_id
LEFT JOIN performance p ON s.employee_id = p.employee_id
GROUP BY d.department_name
ORDER BY "Average score", "Average Salary Amount" DESC;
```

	department_name character varying (30)	Average score numeric	Average Salary Amount numeric
1	Engineering	4.01	80000.00
2	Sales	4.07	82068.97
3	HR	4.11	81818.18
4	Marketing	4.17	77857.14

There appears to be no correlation between performance and salary across departments as marketing department with the highest performance score has the lowest average salary which implies that employees in the department are underpaid for their efforts which in the long run may cause dissatisfaction and employees leaving the company. Sales department with the second lowest score has the highest average salary.

Recommendation

- The company needs to review the salary structure to reflect performance.
- Alternatively, the company can implement an annual award scheme to recognise and reward high performing employees.
- Compare employee salaries to industry standards and market rates to identify roles or levels where pay is below expectations and may affect retention or morale.


THANK YOU!